Performance Testing of a Pelton Wheel Impulse Turbine

Power, torque, and efficiency as functions of speed ratio at three heads and four nozzle openings

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Team	Group A4: Luis Ramirez, Macklin Horner, Zachary Klein, Larimel Rosario, John Wahba	**Equipment**	GUNT HM 290 Pelton turbine (70 mm wheel) with needle nozzle, band brake with torque sensor, speed, pressure, and flow sensors, data acquisition software

PROJECT SUMMARY

A small Pelton wheel was tested at supply pressures of 1.0, 1.5, and 2.0 bar (heads of about 10, 15, and 20 m) with the needle nozzle 100%, 75%, 50%, and 25% open. For each of the 12 settings the brake was tightened in steps from free-running to near stall, and speed, torque, flow, and pressure were logged. Mechanical power rose to a single peak and fell to zero at both stall and runaway, while torque fell almost linearly with speed. Peak efficiencies were 47–54% at speed ratios of 0.27–0.49, and the highest output, 34.9 W, was reached at 2.0 bar with the nozzle fully open. Friction and bucket losses kept both the best and runaway speed ratios below the ideal values of 0.5 and 1.0.

1. Introduction

The Pelton wheel is an impulse turbine used at hydroelectric sites with high head and relatively low flow. Water is accelerated to a free jet in a nozzle and strikes a ring of split buckets, which turn the jet back nearly 180° and take up its momentum. Because the pressure drop occurs entirely in the nozzle, the runner works at atmospheric pressure and its output depends mainly on the jet speed and how fast the buckets move relative to it. The objective of this experiment was to measure the power, torque, and efficiency of a Pelton wheel over its full speed range at several heads and nozzle openings, and to compare the speed ratio at best efficiency with ideal impulse-turbine theory.

2. Theory

Table 1. Performance relations (ρ = 1000 kg/m³, r = 0.035 m, C_v = 0.98)

Quantity	Relation	Notes
Head	$H = p/\rho g + V_1^2/2g$	Inlet velocity head is under 1 mm here, so $H \approx p/\rho g$
Jet velocity	$V_{jet} = C_v\sqrt{(2gH)}$	C_v accounts for nozzle friction
Bucket speed	$V_{wheel} = r\omega = r(2\pi n/60)$	Pitch-circle speed of the buckets
Speed ratio	$\varphi = V_{wheel}/V_{jet}$	Ideal best efficiency at $\varphi = 0.5$; runaway at $\varphi = 1$
Hydraulic power	$E_h = \gamma QH$	Power available in the supply
Mechanical power	$E_b = M\omega$	Brake torque times angular speed
Efficiency	$\eta = E_b/E_h$	Overall turbine efficiency

For an ideal bucket that turns the jet through 180°, the power is proportional to $\varphi(1 - \varphi)$, so it is zero when the wheel is held still ($\varphi = 0$) and when the buckets run as fast as the jet ($\varphi = 1$), with a maximum at $\varphi = 0.5$. Torque is proportional to $(1 - \varphi)$ and so falls linearly with speed. In a real turbine, bucket friction, deflection angles below 180°, windage, and bearing losses lower the peak and shift it to a smaller speed ratio.

3. Procedure

The supply pump was set to 1.0 bar with the nozzle fully open and the brake released, so the wheel ran at its free (runaway) speed. The band brake was then tightened in steps, and at each step the software logged pressure, speed, flow rate, and torque once the readings settled, continuing until the wheel was close to stall. The run was repeated with the nozzle 75%, 50%, and 25% open, and the full set was repeated at 1.5 and 2.0 bar, giving 12 runs of 11–12 points each.

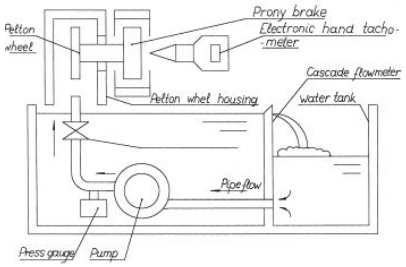


Figure 1. Test circuit: pump, pressure gauge, Pelton wheel, brake, flowmeter

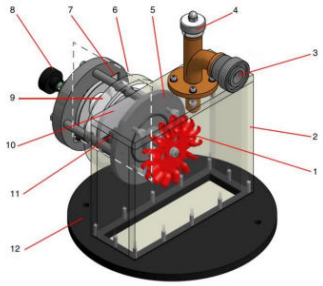


Figure 2. HM 290 Pelton turbine assembly (lab manual)

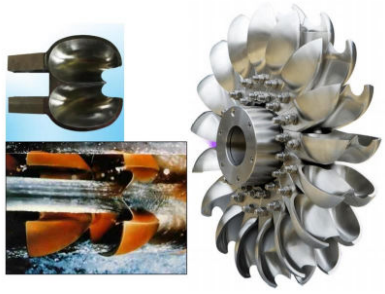


Figure 3. Pelton runner and split buckets

4. Results

Power and torque

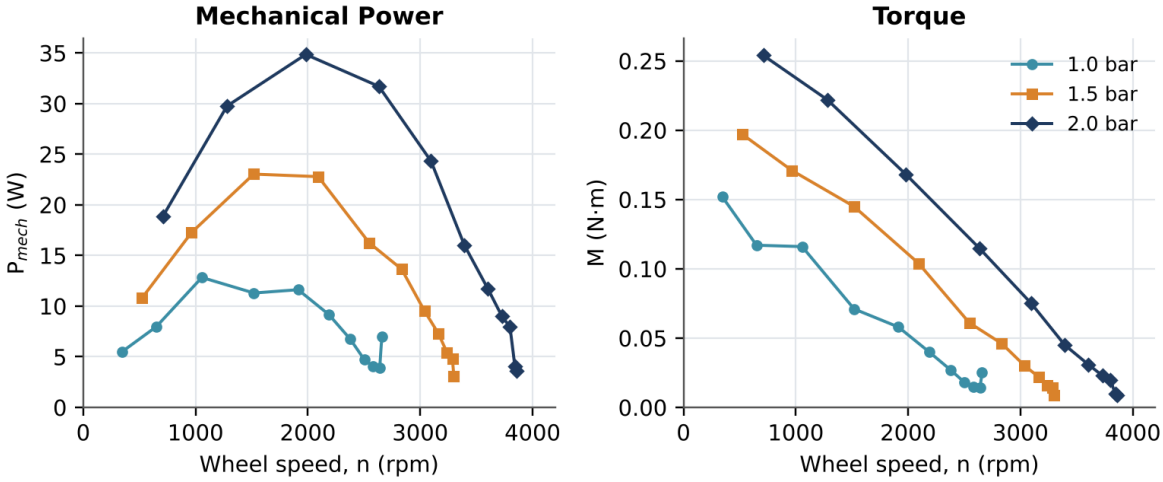


Figure 4. Mechanical power and brake torque vs. wheel speed with the nozzle fully open.

Efficiency

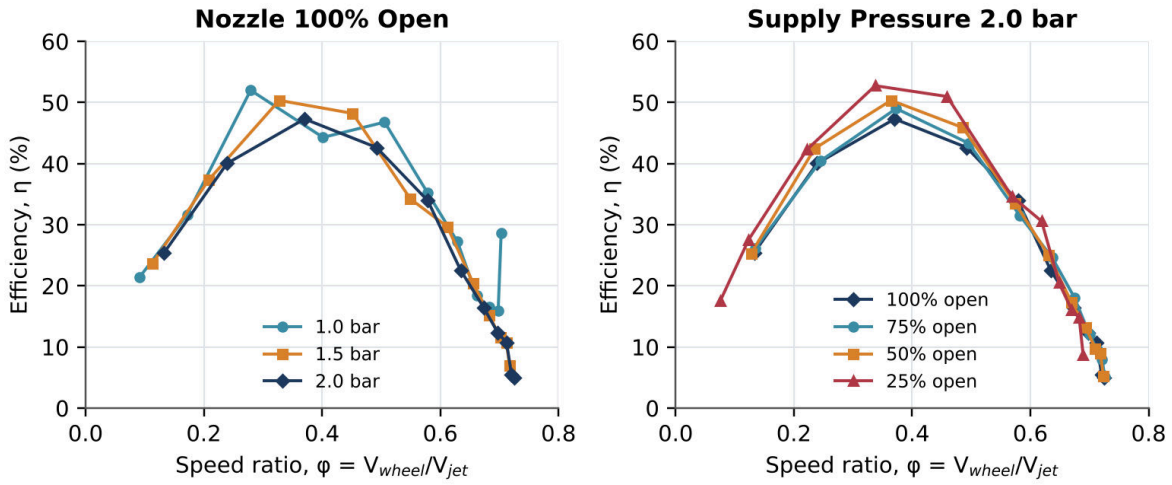


Figure 5. Efficiency vs. speed ratio: three heads with the nozzle fully open (left) and four nozzle openings at 2.0 bar (right).

Table 2. Summary of all 12 runs

p (bar)	Nozzle (%)	Q (L/min)	H (m)	P_{max} (W)	η_{max} (%)	n at η_{max} (rpm)	φ at η_{max}	Runaway n (rpm)	Runaway φ
1.0	100	15.0	10.19	12.8	52.0	1056	0.28	2658	0.70
1.0	75	15.0	10.19	12.9	50.6	1063	0.28	2613	0.71
1.0	50	13.9	10.19	12.8	54.4	1033	0.27	2665	0.70
1.0	25	7.8	10.19	6.8	52.5	1862	0.49	2518	0.66
1.5	100	18.6	15.29	23.0	50.3	1517	0.33	3299	0.72
1.5	75	18.5	15.29	22.8	47.8	1546	0.33	3331	0.72
1.5	50	18.6	15.29	22.9	48.8	1921	0.41	3324	0.72
1.5	25	17.5	15.29	21.9	48.8	2075	0.45	3323	0.72
2.0	100	22.0	20.39	34.9	47.3	1981	0.37	3858	0.73
2.0	75	21.6	20.39	34.5	49.0	1995	0.37	3881	0.73
2.0	50	20.3	20.39	33.1	50.4	1949	0.36	3868	0.72
2.0	25	11.8	20.39	20.8	52.7	1810	0.34	3695	0.69

5. Discussion

Shape of the curves. The measured curves follow impulse-turbine theory. Torque was highest near stall and fell almost linearly as the brake was released, and mechanical power rose to one peak before dropping toward zero at runaway. Raising the supply pressure increased the jet velocity, so both the power and the runaway speed rose with head: with the nozzle fully open the peak power grew from about 13 W at 1.0 bar to 34.9 W at 2.0 bar, and the free-running speed rose from about 2660 to 3860 rpm.

Speed ratio and efficiency. The best efficiencies, 47–54%, occurred at φ ≈ 0.27–0.49, and the wheel ran away at φ ≈ 0.66–0.73 instead of 1.0. Both values sit below the ideal because the small buckets turn the jet through less than 180°, water rubs along the bucket surfaces, and windage and bearing friction absorb part of the output; these losses grow with speed and pull the peak to a lower speed ratio. A 70 mm wheel running on a 10–20 m head is

also far smaller than a commercial unit, where efficiencies near 90% are typical.

Nozzle opening and data quality. At 1.0 and 2.0 bar, closing the needle valve reduced the flow as expected, but at 1.5 bar the flow barely changed (18.3 to 17.7 L/min), which suggests the nozzle setting was not changed much between those runs. One point at 1.0 bar and 25% opening read 80% efficiency with torque rising as the speed increased; it was treated as a sensor glitch and left out of the summary. Torque readings near runaway were only a few N·cm, so small sensor offsets have a large effect on efficiency at high speed ratios.

6. Conclusion

The Pelton wheel behaved as an impulse turbine should: torque decreased linearly with speed, and power and efficiency peaked at an intermediate speed. Peak efficiencies of 47–54% were reached at speed ratios of about 0.27–0.49, below the ideal 0.5 because of bucket friction, imperfect jet deflection, and windage, and the maximum output of 34.9 W came at 2.0 bar with the nozzle fully open. For the best efficiency, a wheel of this size should be loaded so that its bucket speed is roughly one-third to two-fifths of the jet speed.

References

New Jersey Institute of Technology. (n.d.). *Performance test on hydraulic impulse turbine (Pelton wheel)* [ME 405 laboratory manual]. Department of Mechanical and Industrial Engineering.

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